



SIMATS ENGINEERING

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ASSESSMENT TOOL -3

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COMPARATIVE ANALYSIS

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.

ANSWER:-

1. Comparison Between Star Topology and Bus Topology

Introduction

Star topology and bus topology are two widely used network topologies in computer networking. Both are suitable for connecting multiple devices, but they differ in terms of cost, installation, performance, reliability, scalability, and maintenance. Choosing the appropriate topology depends on the organization's budget, network size, and operational requirements. While bus topology is preferred for small, low-cost networks, star topology is commonly used in modern homes, schools, offices, and businesses due to its superior performance and reliability.

Comparison

A star topology is generally more expensive than a bus topology because it requires a central networking device, such as a switch or hub, along with separate cables connecting each device to the central unit. In contrast, a bus topology uses a single backbone cable to which all devices are connected, making it less expensive and easier to install.

In terms of reliability, the star topology performs much better. If one cable or workstation fails, only that particular device is affected, while the remaining devices continue to communicate normally. However, in a bus topology, damage to the main backbone cable can cause the entire network to stop functioning.

Regarding performance, star topology provides faster and more efficient communication because each device has a dedicated connection to the central switch, reducing data collisions and improving network speed. On the other hand, bus topology uses a shared communication medium, which increases the possibility of data collisions and reduces network performance as more devices are connected.

In terms of security, star topology provides better network control because managed switches can implement security features such as VLANs, port security, and traffic monitoring. Bus topology offers limited security since all devices share the same communication medium, making data more vulnerable to unauthorized access.

Advantages and Disadvantages

Star Topology

- High reliability and fault tolerance.
- Easy to install, manage, and troubleshoot.
- Better network performance with fewer data collisions.
- Easily expandable for future growth.
- Higher installation cost due to additional cabling and networking equipment.

Bus Topology

- Low installation cost.
- Requires less cabling and fewer networking devices.
- Simple to set up for small networks.

Conclusion

Overall, the star topology is the better choice for most modern networks because it offers higher reliability, improved performance, stronger security, easier troubleshooting, and better scalability. Although it has a higher installation cost than the bus topology, its long-term benefits make it the preferred option for homes, offices, schools, and business organizations. The bus topology remains a suitable choice only for very small networks with limited budgets and minimal performance requirements.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

ANSWER:-

2. Comparison Between Mesh Topology and Star Topology

Introduction

Mesh topology and star topology are two widely used network topologies designed to provide efficient communication between devices. Although both topologies are reliable, they differ significantly in terms of fault tolerance, cost, performance, scalability, and maintenance. Mesh topology is commonly used in mission-critical environments such as banks, data centers, and military networks, while star topology is widely adopted in offices, schools, and small to medium-sized businesses due to its simplicity and cost-effectiveness.

Comparison

The most significant difference between the two topologies is fault tolerance. A mesh topology provides very high fault tolerance because every device is connected to multiple other devices. This creates multiple communication paths, allowing data to be automatically rerouted if one connection fails. As a result, network communication continues without interruption, making mesh topology highly reliable.

In contrast, a star topology offers moderate fault tolerance. If an individual computer or network cable fails, only that device is affected, and the rest of the network continues to operate normally. However, if the central switch or hub fails, the entire network becomes unavailable because all communication passes through the central device.

In terms of performance, mesh topology provides excellent network performance since multiple communication paths reduce congestion and ensure continuous data transmission. Star topology also offers good performance because each device has a dedicated connection to the switch, but all traffic depends on the capacity of the central networking device.

The cost of implementing a mesh topology is much higher than a star topology. Mesh networks require a large number of cables, network interfaces, and networking devices, making installation and maintenance expensive. Star topology is more economical because it requires only one central switch and individual cables connecting each device to the switch.

Mesh topology also provides better network availability and reliability, making it suitable for organizations where network downtime cannot be tolerated. Star topology, while reliable for everyday business use, is better suited for environments where cost and ease of management are more important than maximum redundancy.

Advantages and Disadvantages

Mesh Topology

- Very high fault tolerance and reliability.
- Multiple communication paths prevent network failure.
- Excellent performance and continuous data transmission.
- Suitable for mission-critical applications.
- High installation and maintenance costs.
- Complex network design and configuration.

Star Topology

- Easy to install, manage, and troubleshoot.
- Lower installation cost compared to mesh topology.
- Good network performance with dedicated connections.
- Easy to expand by adding new devices.
- Failure of the central switch or hub can stop the entire network.

Conclusion

Both mesh topology and star topology have important advantages depending on the network requirements. Mesh topology is the preferred choice for critical environments such as banks, hospitals, and data centers because of its exceptional fault tolerance, redundancy, and reliability.

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

ANSWER : _

Comparison Between Bus Topology and Mesh Topology

Introduction

Bus topology and mesh topology are two different network designs that serve different purposes. Bus topology is known for its simple structure and low implementation cost, making it suitable for small networks with limited budgets. In contrast, mesh topology is designed to provide maximum reliability and fault tolerance, making it ideal for large organizations and mission-critical environments. These topologies differ in terms of installation, cost, performance, reliability, scalability, and maintenance.

Comparison

The main difference between the two topologies is the installation process. A bus topology is simple and easy to install because all devices are connected to a single backbone cable. It requires minimal cabling and networking hardware, allowing the network to be set up quickly and at a low cost.

In comparison, a mesh topology is much more complex to install. Every device is connected to multiple other devices, and in a full mesh topology, each node is directly connected to every other node. As the number of devices increases, the number of required cables, ports, and network interfaces also increases, making installation more difficult, time-consuming, and expensive.

In terms of cost, bus topology is one of the least expensive network topologies because it requires only one backbone cable and minimal hardware. Mesh topology, however, is the most expensive topology due to the large amount of cabling, multiple network ports, and advanced networking equipment required.

Regarding performance, mesh topology provides excellent communication because multiple paths are available for data transmission. This reduces congestion and allows data to continue flowing even if one connection fails. Bus topology relies on a shared communication medium, so performance decreases as more devices are connected, leading to increased data collisions and slower network speeds.

Advantages and Disadvantages

Bus Topology

- Low installation and maintenance cost.
- Simple and quick to install.
- Requires less cabling.
- Suitable for small networks.
- Poor fault tolerance.
- Performance decreases as more devices are added.

Mesh Topology

- Excellent fault tolerance and reliability.
- Multiple communication paths ensure continuous connectivity.
- High network performance.
- Suitable for critical applications such as banks and data centers.
- Very expensive to install and maintain.
- Complex design requiring many cables and network ports.

Conclusion

Overall, bus topology is the best option for small organizations, classrooms, or home networks where cost is the primary concern and network traffic is relatively low. Mesh topology is the preferred choice for organizations that require maximum reliability, high availability, and continuous network operation, even though it involves higher installation costs and greater complexity. The selection between the two depends on the organization's budget, network size, and reliability requirements.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

ANSWER :-

Comparison Between Hybrid Topology and Star Topology

Introduction

Hybrid topology and star topology are widely used in modern computer networks because they provide good performance and flexibility. However, they differ significantly in terms of scalability, flexibility, cost, reliability, and network management. Star topology is commonly used in small and medium-sized organizations, whereas hybrid topology is preferred for large enterprises, universities, hospitals, and organizations with multiple branches that require a highly scalable and reliable network.

Comparison

The primary difference between the two topologies is scalability. A hybrid topology combines two or more network topologies, such as star, mesh, ring, or tree, allowing the network to expand easily as organizational requirements increase. New departments, buildings, or network segments can be added without affecting the existing infrastructure, making hybrid topology highly flexible and future-ready.

A star topology is also scalable but only to a certain extent. Since all devices connect to a central switch or hub, the number of devices that can be added depends on the available ports of the central device. Once the switch reaches its maximum capacity, additional switches or hubs must be installed, increasing the overall cost and complexity.

In terms of flexibility, hybrid topology provides greater freedom because different topologies can be implemented in different parts of the network according to specific requirements. For example, a university may use a mesh topology for the campus backbone, star topology within

buildings, and tree topology for departmental networks. Star topology follows a fixed structure and offers less flexibility in network design.

Regarding reliability, hybrid topology is generally more reliable because it often includes redundant links and multiple communication paths. If one network segment fails, the remaining parts of the network continue to operate normally. Star topology provides good reliability for individual device failures, but if the central switch or hub fails, the entire network becomes unavailable.

Advantages and Disadvantages

Hybrid Topology

- Highly scalable and flexible.
- Supports large and complex network environments.
- High reliability with redundant network paths.
- Easy to expand without affecting existing network segments.
- Suitable for universities, enterprises, hospitals, and large organizations.
- Higher installation, maintenance, and implementation costs.

Star Topology

- Easy to install and manage.
- Good performance with dedicated device connections.
- Lower implementation cost than hybrid topology.
- Easy to troubleshoot and maintain.

Conclusion

Both hybrid topology and star topology are effective networking solutions, but they are designed for different environments. Hybrid topology is the best choice for large organizations that require maximum scalability, flexibility, and reliability because it can easily accommodate future expansion without disrupting the existing network.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

ANSWER :-

Comparison of Network Topologies Based on Maintenance

Introduction

Maintenance is an important factor when selecting a network topology because it affects the time, cost, and effort required to keep a network operating efficiently. Different network topologies have different maintenance requirements depending on their design, complexity, and level of redundancy. Star, mesh, bus, and hybrid topologies each offer unique advantages and challenges in terms of troubleshooting, fault detection, network management, and overall maintenance.

Comparison

Among these topologies, star topology is the easiest to maintain. Since every device is connected to a central switch or hub through an individual cable, network problems can be identified and isolated quickly. If one cable or workstation fails, only that specific device is affected, allowing maintenance to be performed without disrupting the rest of the network. This centralized design also simplifies monitoring and network administration.

In contrast, mesh topology is much more difficult to maintain because every node is connected to multiple other nodes. Although this design provides excellent fault tolerance and high availability, the large number of cables and communication paths makes troubleshooting more complex and time-consuming. Maintaining a mesh network often requires experienced network administrators and specialized monitoring tools.

A bus topology is relatively difficult to maintain compared to a star topology. Since all devices share a single backbone cable, identifying the exact location of a fault can be challenging. If

the backbone cable is damaged or disconnected, the entire network may stop functioning until the problem is located and repaired. As the number of connected devices increases, maintenance becomes even more difficult.

A hybrid topology has moderate maintenance complexity. Because it combines two or more topologies, such as star, mesh, ring, or tree, administrators must monitor multiple network structures simultaneously. While hybrid topology provides flexibility and improved reliability, maintaining different network segments requires careful planning, regular monitoring, and skilled technical personnel.

Advantages and Disadvantages

Star Topology

- Easy to monitor and troubleshoot.
- Faults can be isolated quickly.
- Minimal impact on other devices during maintenance.
- Simple network management.
- Failure of the central switch affects the entire network.

Mesh Topology

- Excellent reliability and fault tolerance.
- Network continues operating even if multiple links fail.
- Complex maintenance due to numerous connections.
- Requires skilled administrators and advanced monitoring tools.
- High maintenance cost.

Bus Topology

- Simple network structure with low installation cost.
- Difficult to locate faults in the backbone cable.
- Entire network may fail if the main cable is damaged.
- Maintenance becomes harder as the network grows.

Hybrid Topology

- Flexible and suitable for large organizations.
- Easier to maintain individual network segments than a full mesh.
- Supports expansion without major disruptions.
- Requires experienced administrators to manage multiple topologies.
- More complex than star topology but simpler than a full mesh network.

Conclusion

From a maintenance perspective, star topology is the easiest and most cost-effective topology to manage because faults can be identified and resolved quickly. Mesh topology offers the highest reliability but requires significant maintenance effort due to its complex structure. Bus topology is inexpensive but difficult to troubleshoot because a single backbone cable failure can affect the entire network. Hybrid topology provides a balanced solution by combining the advantages of different topologies, although it requires skilled administration and careful network management. Therefore, the choice of topology should be based on the organization's size, budget, maintenance capabilities, and reliability requirements.